

Transposition and the Origin in the Orthogonal Euler Circle Frame (Exercise XXX)

Anonymous Author(s)

Abstract

We formalize the transposition (mirror) structure of the orthogonal Euler circle frame — the radius-1 and radius- i circles of the split-complex numbers (Exercise XXVII) — and determine the origin of the frame. Seven theorems, machine-checked with zero sorry: the transposition $(a, b) \mapsto (b, a)$ is an involution; it maps $1 = (1, 0)$ to $i = (0, 1)$; it maps the radius-1 circle $a^2 - b^2 = 1$ to the radius- i circle $a^2 - b^2 = -1$ in both directions; and it fixes the origin $(0, 0)$. The natural-number lattice of each circle is a single point: the radius-1 circle contains exactly $(1, 0)$ and the radius- i circle exactly $(0, 1)$ — the two units are transposes of each other, and the earlier observation that lattice observation degenerates to $\{1, i\}$ is made precise. The origin of the frame is $(0, 0)$: the fixed point of the transposition and the common center of the two conjugate hyperbolas. It is neither 1 nor i .

Keywords: Lean, formalization, split-complex, transposition, origin, 0pat

1 The frame and the question

The orthogonal Euler circle frame (Exercise XXVII) consists of the radius-1 circle $a^2 - b^2 = 1$ and the radius- i circle $a^2 - b^2 = -1$ in the split-complex plane $a + bj$, $j^2 = +1$. The frame was built to observe the prime axis through two orthogonal circles with different units (1 and i). This exercise formalizes the transposition structure of the frame and answers a positional question: in this frame, where is the origin?

2 The theorems

Transposition (`transpose_involutive`, `transpose_maps_one_to_i`).

The map $(a, b) \mapsto (b, a)$ exchanges the coordinates. It is an involution (applying it twice returns the point), and it maps the unit of the radius-1 circle to the unit of the radius- i circle: $1 = (1, 0) \mapsto (0, 1) = i$.

The circles are transposes (`transpose_maps_unit_to_imaginary`, `transpose_maps_imaginary_to_unit`).

A point (a, b) lies on the radius-1 circle exactly when its transpose (b, a) lies on the radius- i circle, in both directions: $a^2 - b^2 = 1 \leftrightarrow b^2 - a^2 = -1$. The two circles are mirror images under the transposition.

Lattice uniqueness (`lattice_unique_unit`, `lattice_unique_imaginary`).

Under the constraint that both coordinates are natural numbers, each circle contains exactly one point: the radius-1 circle contains only $(1, 0) = 1$, and the radius- i circle contains only $(0, 1) = i$. The proofs are elementary: for $(a, b) \in \mathbb{N}^2$, $a^2 - b^2 = 1$ forces $b = 0$ (otherwise the difference is at least 3) and then $a = 1$; symmetrically for $b^2 - a^2 = 1$. This makes precise the earlier observation that lattice observation of the frame degenerates to the two units $\{1, i\}$: each circle contributes exactly its own unit, and the two units are transposes of each other.

The origin (`transpose_origin_fixed`). The transposition fixes the origin: $(0, 0) \mapsto (0, 0)$. The origin is the common center of the two conjugate hyperbolas $a^2 - b^2 = \pm 1$ — the point about which the two circles are symmetric under the transposition $(a, b) \leftrightarrow (b, a)$ combined with negation. It is neither 1 nor i : those are the units of the two circles, the unique lattice points, and the transposition pairs them; the origin is their fixed point.

3 Honest boundary and position

The content is elementary algebra (coordinate transposition, lattice equations in $\mathbb{N}$); the value is positional: the orthogonal Euler circle frame now has its transposition structure formalized and its origin determined. The origin of the frame is $(0, 0)$ — the split zero and the common center of the two circles. The result concerns the frame, not the Riemann zeta function: no claim is made about zeros, and the analysis of ζ remains the open span of the bridge.

4 Formalization

The proof is a single Lean file (`proof.lean`, 185 lines), importing only `Mathlib.Data.Real.Basic` and basic tactics. The split-complex structure (30 lines, as in Exercises XXVII–XXVIII) is self-contained. Seven theorems, compiled with Lean 4.32.2 / `mathlib` v4.32.2, zero errors, zero sorry.

5 Conclusion

Theorem	Statement	Proof
<code>transpose_involutive</code>	$T(Tz) = z$	The orthogonal Euler circle frame is now fully determined:
<code>transpose_maps_one_to_i</code>	$T(1) = i$	its transposition structure is formalized (involution, $1 \leftrightarrow i$,
<code>transpose_origin_fixed</code>	$T(0) = 0$	circles as mirrors), its lattice observation degenerates to the
<code>transpose_maps_unit_to_imaginary</code>	$\ z\ ^2 = 1 \Rightarrow \ Tz\ ^2 = -1$	two transposed units $\{1, i\}$, and its origin is $(0, 0)$ — the com-
<code>transpose_maps_imaginary_to_unit</code>	$\ z\ ^2 = -1 \Rightarrow \ Tz\ ^2 = 1$	mon center of the two circles, fixed by the transposition, nei-
<code>lattice_unique_unit</code>	$a^2 - b^2 = 1 \Rightarrow (a, b) = (1, 0)$	ther 1 nor i . The frame is complete; the analysis of ζ remains
<code>lattice_unique_imaginary</code>	$b^2 - a^2 = 1 \Rightarrow (a, b) = (0, 1)$	the open span of the bridge.

References